

Solutions: Additional Exercises 2

Partial Derivatives & Differentiability

Exercise 1: First-order partial derivatives

- (a) $z = x^3 + 3x^2y - y^3$: $z_x = 3x^2 + 6xy$, $z_y = 3x^2 - 3y^2$.
- (b) $z = \ln(x^2 + y^2)$: $z_x = \frac{2x}{x^2+y^2}$, $z_y = \frac{2y}{x^2+y^2}$.
- (c) $z = \arctan(y/x)$: $z_x = \frac{1}{1+(y/x)^2} \cdot \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2+y^2}$, $z_y = \frac{x}{x^2+y^2}$.
- (d) $z = \cos(ax - by)$: $z_x = -a \sin(ax - by)$, $z_y = b \sin(ax - by)$.
- (e) $u = xe^{-yx}$: $u_x = e^{-yx} + x(-y)e^{-yx} = e^{-yx}(1 - xy)$, $u_y = x \cdot (-x)e^{-yx} = -x^2e^{-yx}$.
- (f) $u = y/x + z/y + x/z$: $u_x = -y/x^2 + 1/z$, $u_y = 1/x - z/y^2$, $u_z = 1/y - x/z^2$.
- (g) $z = xy/(x - y)$: $z_x = \frac{y(x-y)-xy}{(x-y)^2} = \frac{-y^2}{(x-y)^2}$, $z_y = \frac{x(x-y)+xy}{(x-y)^2} = \frac{x^2}{(x-y)^2}$.
- (h) $u = \arcsin(y/x)$: $u_x = \frac{1}{\sqrt{1-(y/x)^2}} \cdot \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2\sqrt{1-y^2/x^2}} = \frac{-y}{x\sqrt{x^2-y^2}}$, $u_y = \frac{1}{x\sqrt{1-y^2/x^2}} = \frac{1}{\sqrt{x^2-y^2}}$.
- (i) $u = \sin^2(x + y) - \sin^2 x - \sin^2 y$. Using $\sin^2 \alpha = \frac{1-\cos 2\alpha}{2}$: $u = \frac{1-\cos 2(x+y)}{2} - \frac{1-\cos 2x}{2} - \frac{1-\cos 2y}{2} = \frac{1-\cos 2(x+y)}{2} - \frac{1-\cos 2x}{2} - \frac{1-\cos 2y}{2}$. $u_x = -\sin(2x+2y) + \sin 2x = \sin 2x - \sin(2x+2y)$. Using sum-to-product: $u = \sin(x+y+x) \cdot \sin(-(y))$... alternatively: $u_x = 2 \sin(x+y) \cos(x+y) - 2 \sin x \cos x = \sin(2x+2y) - \sin 2x$. Wait, let's be careful: $\frac{d}{dx} \sin^2(x+y) = 2 \sin(x+y) \cos(x+y) \cdot 1 = \sin(2x+2y)$. $\frac{d}{dx} \sin^2 x = 2 \sin x \cos x = \sin 2x$. $\frac{d}{dx} \sin^2 y = 0$. So $u_x = \sin(2x+2y) - \sin 2x$. Similarly $u_y = \sin(2x+2y) - \sin 2y$. One can verify $u = \frac{1}{2}[\cos 2x + \cos 2y - \cos(2x+2y) - 1] = -\frac{1}{2} \cos 2y \cdot 1$... Let's just leave as: $u_x = \sin(2x+2y) - \sin 2x$, $u_y = \sin(2x+2y) - \sin 2y$.
- (j) $z = \ln(\sqrt[3]{1/x} - \sqrt[3]{1/y}) = \ln(x^{-1/3} - y^{-1/3})$. $z_x = \frac{-\frac{1}{3}x^{-4/3}}{x^{-1/3}-y^{-1/3}} = \frac{-1}{3x^{4/3}(x^{-1/3}-y^{-1/3})} = \frac{-1}{3x(1-x^{1/3}y^{-1/3})}$ (multiply numerator/denominator by $x^{1/3}$) $= \frac{-1}{3(x-x^{2/3}y^{-1/3})}$... Let $A = x^{-1/3} - y^{-1/3}$. $z_x = \frac{-\frac{1}{3}x^{-4/3}}{A}$, $z_y = \frac{\frac{1}{3}y^{-4/3}}{A}$.

Exercise 2: Partial derivatives at (0,0) from definition

Method: $f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x}$.

- (a) $f(x, 0) = \frac{x^3}{x^2} = x$ for $x \neq 0$, $f(0, 0) = 0$. $f_x(0, 0) = \lim_{x \rightarrow 0} \frac{x-0}{x} = 1$. By symmetry (swap x and y): $f_y(0, 0) = 1$.
- (b) $f(x, 0) = \frac{0}{x^2} = 0$, so $f_x(0, 0) = \lim_{x \rightarrow 0} \frac{0}{x} = 0$. Similarly $f(0, y) = 0$, so $f_y(0, 0) = 0$.
- (c) $f(x, 0) = 0$ for $x \neq 0$. $f_x(0, 0) = \lim_{x \rightarrow 0} \frac{0}{x} = 0$. Similarly $f_y(0, 0) = 0$.
- (d) $f(x, 0)$: at $y = 0$ the formula gives $0 + 0 = 0$ (second branch when $y = 0$). $f_x(0, 0) = \lim_{x \rightarrow 0} \frac{f(x, 0) - 0}{x} = \lim_{x \rightarrow 0} \frac{0}{x} = 0$. Similarly $f_y(0, 0) = 0$.
- (e) $f(x, 0) = 0$. $f_x(0, 0) = 0$. $f(0, y) = 0$. $f_y(0, 0) = 0$. So $f_x(0, 0) = f_y(0, 0) = 0$.

Exercise 3: Verify identities

- (a) $z = \ln(\sqrt{x} + \sqrt{y})$: $z_x = \frac{1/(2\sqrt{x})}{\sqrt{x} + \sqrt{y}}$, $z_y = \frac{1/(2\sqrt{y})}{\sqrt{x} + \sqrt{y}}$. $xz_x + yz_y = \frac{x \cdot \frac{1}{2\sqrt{x}} + y \cdot \frac{1}{2\sqrt{y}}}{\sqrt{x} + \sqrt{y}} = \frac{\frac{\sqrt{x}}{2} + \frac{\sqrt{y}}{2}}{\sqrt{x} + \sqrt{y}} = \frac{\frac{\sqrt{x} + \sqrt{y}}{2}}{\sqrt{x} + \sqrt{y}} = \frac{1}{2}$. ✓
- (b) $u = e^{x/t^2}$: $u_x = \frac{1}{t^2} e^{x/t^2}$, $u_t = e^{x/t^2} \cdot \left(-\frac{2x}{t^3}\right)$. $2xu_x + tu_t = 2x \cdot \frac{e^{x/t^2}}{t^2} + t \cdot \left(-\frac{2x}{t^3}\right) e^{x/t^2} = \frac{2x}{t^2} e^{x/t^2} - \frac{2x}{t^2} e^{x/t^2} = 0$. ✓
- (c) $u = f(x^2 + y^2 + z^2)$. Let $r^2 = x^2 + y^2 + z^2$. $u_x = f'(r^2) \cdot 2x$, $u_y = f'(r^2) \cdot 2y$. $yu_x - xu_y = y \cdot 2x f'(r^2) - x \cdot 2y f'(r^2) = 0$. ✓
- (d) $u = \sqrt{x^2 + y^2 + z^2}$. $u_x = \frac{x}{u}$, $u_y = \frac{y}{u}$, $u_z = \frac{z}{u}$. $u_x^2 + u_y^2 + u_z^2 = \frac{x^2 + y^2 + z^2}{u^2} = \frac{u^2}{u^2} = 1$. ✓
- (e) $u = f(x/y, y/z)$. Let $s = x/y$, $t = y/z$. $u_x = f_s \cdot (1/y)$, $u_y = f_s \cdot (-x/y^2) + f_t \cdot (1/z)$, $u_z = f_t \cdot (-y/z^2)$. $xu_x + yu_y + zu_z = \frac{x}{y} f_s + y \left(-\frac{x}{y^2} f_s + \frac{1}{z} f_t\right) + z \left(-\frac{y}{z^2} f_t\right) = \frac{x}{y} f_s - \frac{x}{y} f_s + \frac{y}{z} f_t - \frac{y}{z} f_t = 0$. ✓

Exercise 4: Differentiability at $(0, 0)$

- (a) From Ex. 2(a): $f_x(0, 0) = f_y(0, 0) = 1$. Test: $\frac{\Delta f - \Delta x - \Delta y}{|\Delta \rho|}$ where $\Delta \rho = \sqrt{(\Delta x)^2 + (\Delta y)^2}$. With $\Delta x = r \cos \theta$, $\Delta y = r \sin \theta$: $\Delta f = r(\cos^3 \theta + \sin^3 \theta)$ (from polar calculation), linear part $= r \cos \theta + r \sin \theta$. Remainder: $r(\cos^3 \theta + \sin^3 \theta - \cos \theta - \sin \theta) = r(-\cos \theta \sin^2 \theta - \sin \theta \cos^2 \theta) = -r \sin \theta \cos \theta (\cos \theta + \sin \theta)$. Divide by r : $-\sin \theta \cos \theta (\cos \theta + \sin \theta)$, which does *not* go to 0 (depends on θ). **Not differentiable at $(0, 0)$.**
- (b) From definition: $f(x, 0) = x^2 \cos(\pi/x^2)$ for $x \neq 0$. $f_x(0, 0) = \lim_{x \rightarrow 0} \frac{x^2 \cos(\pi/x^2) - 0}{x} = \lim_{x \rightarrow 0} x \cos(\pi/x^2) = 0$ (squeeze). Similarly $f_y(0, 0) = 0$. Differentiability test: $\frac{f(r \cos \theta, r \sin \theta) - 0}{r} = \frac{r^2 \cos(\pi/r^2)}{r} = r \cos(\pi/r^2) \rightarrow 0$ as $r \rightarrow 0$. **Differentiable at $(0, 0)$.**
- (c) (a) $f_x(0, 0) = \lim_{x \rightarrow 0} \frac{\sqrt[3]{x \cdot 0} - 0}{x} = 0$. Similarly $f_y(0, 0) = 0$.
- (b) Away from axes: $f_x = \frac{1}{3}(xy)^{-2/3} \cdot y = \frac{y^{1/3}}{3x^{2/3}}$. As $(x, y) \rightarrow (0, 0)$ along $y = x$: $f_x = \frac{x^{1/3}}{3x^{2/3}} = \frac{1}{3x^{1/3}} \rightarrow \infty$. So f_x is *not* continuous at $(0, 0)$.
- (c) Differentiability: $\frac{f(r \cos \theta, r \sin \theta) - 0}{r} = \frac{(r^2 \cos \theta \sin \theta)^{1/3}}{r} = \frac{r^{2/3} (\cos \theta \sin \theta)^{1/3}}{r} = \frac{(\cos \theta \sin \theta)^{1/3}}{r^{1/3}} \rightarrow \infty$ when $\cos \theta \sin \theta \neq 0$. **Not differentiable at $(0, 0)$.**

- (d) $f(x, 0) = 0$, so $f_x(0, 0) = 0$. Similarly $f_y(0, 0) = 0$. Away from origin: $f_x = \frac{2xy(x^2+y^2)-x^2y \cdot 2x}{(x^2+y^2)^2} = \frac{2xy^3}{(x^2+y^2)^2}$. Along $y = x$: $f_x = \frac{2x^4}{4x^4} = \frac{1}{2} \not\rightarrow 0 = f_x(0, 0)$. So f_x not continuous. But differentiability: $\frac{f(r \cos \theta, r \sin \theta)}{r} = \frac{r^3 \cos^2 \theta \sin \theta}{r^2 \cdot r} = \cos^2 \theta \sin \theta$, which depends on θ . **Not differentiable at $(0, 0)$.**

Exercise 5: Tangent planes and normal lines

- (a) $f = x^2 + y^2$, $f_x = 2x$, $f_y = 2y$. At $(1, 1)$: $f_x = 2$, $f_y = 2$, $f = 2$. Tangent plane: $z - 2 = 2(x - 1) + 2(y - 1)$, i.e. $z = 2x + 2y - 2$. Normal direction: $(f_x, f_y, -1) = (2, 2, -1)$. Normal line: $\frac{x-1}{2} = \frac{y-1}{2} = \frac{z-2}{-1}$.
- (b) $f = e^x \sin y$, $f_x = e^x \sin y$, $f_y = e^x \cos y$. At $(0, \pi/2)$: $f = 1$, $f_x = \sin(\pi/2) = 1$, $f_y = \cos(\pi/2) = 0$. Tangent plane: $z - 1 = 1 \cdot (x - 0) + 0 \cdot (y - \pi/2)$, i.e. $z = x + 1$.
- (c) $f = \ln(x^2 + y^2)$. At $(1, 0)$: $f = 0$, $f_x = 2x/(x^2 + y^2)|_{(1,0)} = 2$, $f_y = 0$. Tangent plane: $z = 2(x - 1)$, i.e. $z = 2x - 2$.
- (d) $f = \sqrt{x^2 + y^2}$. At $(3, 4)$: $f = 5$, $f_x = 3/5$, $f_y = 4/5$. Tangent plane: $z - 5 = \frac{3}{5}(x - 3) + \frac{4}{5}(y - 4)$, i.e. $z = \frac{3}{5}x + \frac{4}{5}y$.
- (e) $F(x, y, z) = x^2 + y^2 + z^2 - 14$. $\nabla F = (2x, 2y, 2z)$. At $(1, 2, 3)$: $\nabla F = (2, 4, 6)$. Tangent plane: $2(x - 1) + 4(y - 2) + 6(z - 3) = 0$, i.e. $x + 2y + 3z = 14$. Normal line: $(x, y, z) = (1, 2, 3) + t(2, 4, 6)$.
- (f) $f = x^2 - xy + y^2 + 1$. $f_x = 2x - y$, $f_y = -x + 2y$. Plane $2x + 4y - z = 5$ has normal $(2, 4, -1)$. Tangent plane normal is $(f_x, f_y, -1)$. For parallel planes, need $f_x/2 = f_y/4 = (-1)/(-1) = 1$, so $f_x = 2$, $f_y = 4$. $2x - y = 2$ and $-x + 2y = 4$. From first: $y = 2x - 2$. Substitute: $-x + 2(2x - 2) = 4 \Rightarrow 3x = 8 \Rightarrow x = 8/3$, $y = 10/3$. Point on surface: $z = f(8/3, 10/3) = 64/9 - 80/9 + 100/9 + 1 = 84/9 + 1 = 93/9 + 9/9 = \frac{102}{9} = \frac{34}{3}$. **Point:** $(\frac{8}{3}, \frac{10}{3}, \frac{34}{3})$.